Self-Audit: public mathematical/physics work of Arvin Hampton (539 Labs)

Referee-style adversarial audit, prepared 19 Oct 2026 (America/Chicago). Read-only: nothing was pushed, posted, filed or sent. Every verification script and its output is in `scripts/`. Citations take the form `record-id:line`, which means the line in `zenodo/txt/<id>_*.txt` (pdftotext -layout), or `file:line`, or a URL.

1. Scope and inventory

- **Zenodo.** I ran 8 API queries: creators.name "Hampton, Arvin", "Arvin Hampton", "539 Labs", "Residual Ternary Algebra", HQH-539-512, "17 Theorems of Entanglement", S2-11DM2ET-X, and "Bridge C" Hampton.
  - After filtering to records whose creators include Arvin Hampton: **48 records, 48 PDFs, 502 pages**, dated 20 Nov 2025 to 28 Sep 2026.
  - The index is `zenodo/index.tsv` and the text extracts are in `zenodo/txt/`.
  - Record 23445098 (Parity Dominance Theorem, 18 Oct 2026) has a .docx only. It is not audited.
- **GitHub.** ArvinHampton/539-Labs-Master, shallow read-only clone in `repo/` (HEAD cb11076, 19 Oct 2026 09:48 CDT).
  - 2443 files; 1399 notes under `docs/S2-11DM2ET-X-Axiomatic-Book/`.
  - The reference `hqh539_core.py` named in the docs is not in the public tree.
- **Local manuscript.** Bridge C Clean v2: 70 pp, `heavy-daily/2026-10-19-Bridge-C-Clean-v2/`, 26 tex sections, 2557 lines. Also the Certified-SRho, Certified-SRho-Check and HighRAM-Items folders.
- **X.** @539Labs (Arvin Hampton, id 1961794934307999744). I searched 21 recent posts (15–19 Oct 2026). The handle @539LabsPQC listed on the website does not exist on X.
- **Website.** `https://539labs.org` (home page, fetched 19 Oct 2026).

Coverage is uneven, and this is deliberate. Bridge C was spot-checked in depth. Physics papers were sampled at their headline claims. I did not read all 121 pp of 18442276 or all 1399 repo notes. Points I could not substantiate were dropped.

2. What holds up (verified)

Bridge C Clean v2

Script: `scripts/verify_bridgeC.py`, output in `.out`; exact SymPy arithmetic. Every check below **passed**.

- The norm map holds: N(beta)/400 = q(nu) for beta = (2X+7Y)+Y*sqrt35 and (X,Y) = 10 nu (s04_rq.tex:50).
  - q(v+a) = Q7(v) - 7/20.
  - Nm a = 20 Q7(v) - 7, using Nm d' = 4*5.
  - The first norms are 13, 73, 133, 153, 173, as stated (s04_rq.tex:57).
- The unit eps = 6+sqrt35 has norm +1. beta(gamma nu) = eps beta(nu) holds exactly, with gamma = rho2 rho1 = [[-1,-7],[2,13]] (s03_main.tex:26, s04_rq.tex:37).
  - rho1 and rho2 preserve the Gram matrix [[2,7],[7,7]] (discriminant 35).
  - eps ≡ 1 mod p2 and mod p5.
- The constants agree to all printed digits:
  - I = log(eps)/(2 sqrt35) = 0.2094198203147146314730... (s01_intro.tex:100);
  - 2 log eps = 4.95578;

  - a_C = 0.8291426171 (s03_main.tex:38).

- For the lattice <2>+<-70> the discriminant group has order 140, matching the 140-label family.
  - The number of ±-orbits is 72 = dim Fix(P) (s08_split.tex:73).
  - The Minkowski bound is 5.916 (s04_rq.tex:63).

- **Not independently verified:** dim V = 42 = 36+6, the invariant counts at s = 121 and 169, and the class-group structures. These need the author's PARI/Weil-rep code. I re-derived none of them.

- **Computer-assisted Category A.** The dim S(rho) certificate (`Certified-SRho/certify.py:1-11`) really uses python-flint arb/acb balls at 128 bits. A separate re-implementation (`Certified-SRho-Check/CHECK_INEQUALITY_2026-10-19.md`) reproduces two of the four sectors with margins ≥ 1e12.
  - That re-implementation keeps its failed attempts on record.
  - Its K0 tail bound K0(x) ≤ sqrt(pi/2x) e^-x is proved, not sampled.
  - This is the correct standard for "Category A (computer-assisted)" (s01_intro.tex:76).
  - Caveat: both runs are internal to the project, and sectors B1X+ and A3X- were not re-run independently.

- **Honest scoping.** The manuscript separates Category A from Category B. The repo README (README.md:42) states plainly that the raw T3 map has "no natural emergence of exactly 539 steps". The HQH formal-games note (HQH539_Formal_Games_and_Hard_Problem_Pi.md:100) admits that no security reduction exists. X post 2112001852963258497 discloses that the Ramanujan Journal rejected the paper.

Residual Ternary Algebra (22143607)

The elementary packaging arithmetic is correct: $3\cdot68 + 5\cdot67 = 539$, $18 + 521 = 539$, and $539 = 7^2\cdot11$. The g_s series is defined correctly. For s = 1 it is, as stated, Ramanujan's f_0 shape. The paper's own Supplement §1 says it does not improve on the classical theory, which is an appropriate disclaimer.

3. Errors found (recomputed)

Scripts: `scripts/verify_zenodo_numerics.py`, `verify_17thm.py`, `verify_T3.py` and `verify_HQH_entropy.py`, each with a `.out` file.

#	Claim (cite)	Recomputed	Severity
E1	s0 = floor(sqrt(hbar c/Lambda)) = 1 616 033 987 (17703994:39)	$\hbar c/\Lambda$ is not dimensionless. Its SI value gives 1.69e13, not 1.616e9.	High
E2	$(61^2\cdot55)$^3/2·3^1/6 = 58 481 "exact integer" (17703994:60)	1.112e8, not an integer	High
E3	58 481 × 539.9 s = tropical year "to 13 decimals" (17703994:60ff)	31 573 891.9 s against 31 556 925.2 s, off by 16 967 s (4.7 h)	High
E4	kappa = $(61^2\cdot55)$^-1/3 = 0.181000 (17703994:90)	0.016969	High
E5	$539 = 55 + 55\varphi^7$ - 12·55 + 484 (17703994:81)	1475.89	High
E6	40.00 Hz = 539 × 25 fs (17703994:78)	539×25 fs = 1.35e-11 s, which is 74 GHz	High
E7	The 55-step orbit from s0 with 61 divisions by 3 (17703994:45-51)	The stated map sends s0 to 0 in 21 steps, a fixed point. Each step divides by 3 exactly once, so a 55-step orbit cannot contain 61 divisions.	High
E8	δa_μ prefactor = 2.51e-9 (17703994:25,97)	The stated formula $(61/55)^3\cdot3$^1/6·kappa gives 2.78e-2	High
E9	τ_settle = $\hbar$/E_bind ≈ 22.5 **attoseconds** (18442276:813, Thm 1)	$\hbar$/29.3 meV = 2.25e-14 s = **22.5 fs**, a factor-1000 unit error. The headline "234 as" match "within 1%" of TDSE (18442276:817) therefore fails: the paper's own chain gives about 517 fs.	High
E10	N_flux = floor(e^3×35) = 4880 (21514429:25)	floor($e^3\cdot35$) = 702. Note 4880 ≈ $e^3\cdot243$.	Medium

#	Claim (cite)	Recomputed	Severity
E11	HQCC: every seed terminates in 539±1 steps (18603243:16,53; 20709375:32,73; 21514429:15; 539labs.org)	False for both published maps. The 17703994 map takes about $\log_3$ n steps. The canonical T3 takes 62–147 steps for 60-bit seeds (verify_T3.out). The repo itself concedes this (README.md:42).	High
E12	$\alpha = 1/137$ "exact at tree level" (18603243:85)	N_eff = 137 is asserted with "explicit enumeration in the project files", and no enumeration is given. 1/137 differs from CODATA α by 2.6e-4 (relative).	High
E13	HQH-539-512 "pre-image $\geq$ 3^539 (≈854-bit), collision $\geq$ 3^270 (≈428-bit)" (539labs.org)	Impossible for any function with a 512-bit output: the generic bounds are 2^512 and 2^256. Measured: after 539 T3 steps the 512-bit state shrinks to 131–264 bits (mean 196). T3 is not injective (T3(5) = T3(9) = 3), and the image of T3^50 on [0, 3^12) holds 160 points. The body therefore lowers the generic collision bound below SHA3-512's 2^256, to about 2^n/2 for an n ≈ 200–260-bit state (empirical, not a proof).	Critical (product claim)

4. Overclaims

- **Millennium "Resolutions"** (2 pp each, 23–24 Dec 2025): Navier–Stokes (18040014:13), Yang–Mills mass gap (18040055:12), **"Theorem: P = NP"** (18040068:38) and Hodge (18049961:50).
  - The "proofs" rest on undefined "debit" and "multiversal leakage across D2-branes", with numerical parameters such as β_PBH = 0.18 and "0.25".
  - None of them contains a mathematical argument a referee could check.
  - The P = NP claim also contradicts the HQH-539 security claims made by the same author.
  - Severity: **Critical**; these should be withdrawn or retitled.
- **"Exact validation... across the entire periodic table"** (20617775, 20709375). A free factor Ω_11D "absorbs all scaling" (20617775:87). That makes the fit unfalsifiable, so "parameter-free" and "exact" are false by construction.
  - "χ^2/dof < 0.82" is quoted without data, degrees of freedom or a likelihood. The same number is reused for the Hubble tension (17959689:2).
- **"Parameter-free derivation of α"** (18603243): see E12.
  - Muon g-2 "perfect agreement" (17703994:97): see E8. The quoted 4.8σ is a tension figure, not the model's significance.
- **Website.** It calls the work "Theory of Everything" and the HQCC Theorem "proves" 539±1 steps (contradicts README.md:42). It says the T3 decomposition is "computationally irreducible under all polynomial-time reductions, classical or quantum" (contradicts HQH539_Formal_Games... md:100, "no reduction"). It claims a qutrit "raw error rate < 1e-20 per gate", with no data. It says security is "supported by" the Ramanujan/K3/RTA papers, but none of them contains a cryptographic argument.
- **X.** "Proof of how Ramanujan's work was the key to the Unification of Physics" (x.com/539Labs/status/2112193351931830629). "Resolution of all anomalies and paradoxes, and parameter-free recovery of the entirety of the standard [model]" (.../2110819448542384223). False vacuum decay an "absolute structural impossibility" (.../2111889267979153509). None of these is supported by any paper in the inventory.

5. Unsubstantiated, or standard results relabelled

- **Bridge C.** Much of it is classical theory under project names. This is not wrong, and the manuscript mostly cites the sources, but the novelty claims should be narrowed. The classical pieces are:
  - Hecke/Shintani partial zeta values of a ray class of $\mathbb{Q}(\sqrt{35})$;

  - Zwegers indefinite theta series;
  - Siegel theta lifts and Weil representations of a discriminant form;
  - Lewis–Zagier period functions;
  - Baker's theorem.

- **Bridge C Category A labels stronger than their support:**
  - s14_hs.tex:83: "two independent symbolic routes agree to 170 digits at 8 values of s" is evidence, not proof, unless both routes are exact symbolic identities in s.
  - s14_hs.tex:89: the "R + S$\sqrt{\Delta}$ split is interpolated and re-verified at 10 held-out values". Interpolation proves an identity only with an a-priori degree bound, which should be stated, or the label should be Category B.
  - s08_split.tex:90: "Category A given the above". The dependency should be named.
- **17 Theorems.** Theorems 7–17 are "invariance/independence" statements of a formula. They follow from parameters not appearing in it, so they are tautologies, not theorems about entanglement. Theorem 6 (no superluminal signalling) is the standard no-signalling theorem of quantum mechanics.
- **RTA.** "Discrete bridge from E8 to the fifth order" and "fifth-order unification" are titles only. The verified content is packaging integers and the standard f_0 form.
- **HQH-539.** The construction is SHA3-512 $\to$ 539 iterations of a lossy piecewise-linear map $\to$ SHA3-512. Any security it has comes from SHA3. No reduction exists, by the author's own repo note.

6. Recommendations, ranked by severity

- **Critical.** Withdraw or retitle the four Millennium "Resolution" records (18040014, 18040055, 18040068, 18049961). Remove "P = NP" in particular.
- **Critical.** Remove the $\geq$3^539 / 854-bit / 428-bit security figures from 539labs.org. They exceed the generic limits of a 512-bit output. State that the T3 body is lossy (E13) and that security rests on SHA3-512 alone. Remove "irreducible under all polynomial-time reductions".
- **High.** Publish errata for 17703994 (E1–E8), 18442276 Thm 1 (E9: as$\to$fs, and withdraw the "within 1% of TDSE" claim), 21514429 (E10) and 18603243 (E12).
- **High.** Retract the "HQCC: all seeds terminate in 539$\pm$1 steps" statement everywhere (Zenodo, website). The repo already says it is false.
- **High.** Drop "parameter-free/exact" wherever a fitted Ω_11D or similar factor is present. Publish data and the χ^2 computation, or remove the χ^2/dof figures.
- **Medium.** Align the X/website language ("proof of unification", "resolution of all anomalies", "ToE") with Category B status.
- **Medium.** Bridge C: relabel s14_hs.tex:83, :89 as Category B, or add degree bounds or a symbolic proof. Get an external re-run of the remaining two S(ρ) sectors. Release the PARI scripts for dim V = 42 and the invariant counts.
- **Low.** Narrow the Bridge C novelty claims to the specific new identities, crediting the classical frameworks listed in §5.

Files

- Scripts: `scripts/verify_bridgeC.py`, `verify_zenodo_numerics.py`, `verify_17thm.py`, `verify_T3.py`, `verify_HQH_entropy.py`, each with a `.out` file.
- Zenodo: `zenodo/index.tsv` and the PDFs; text extracts in `zenodo/txt/`.
- Repo: `repo/`.